Prasanna Govindarajulu Sairam

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EDUCATION

University at Buffalo, School of Engineering and Applied Sciences - Expected Spring 2027
Bachelor of Science in Computer Science
Relevant Coursework: *Object-Oriented Programming & Data Structures, Algorithms & Complexity, Systems Programming, Computer Organization, Web Applications, Probability & Statistics*

PROJECTS

Backend Web Server | Python, MongoDB, WebSockets, bcryptSept – Dec 2025

- Built a custom HTTP server in Python from scratch, handling request parsing, routing, and response generation without using high-level web frameworks.
- Designed RESTful API endpoints supporting user authentication, real-time chat messaging, and file uploads.
- Implemented secure session management with hashed cookies and bcrypt password hashing to protect user credentials.
- Integrated MongoDB to store users, chat history, and media, and added WebSocket handshakes to enable real-time messaging.

OCaml Interpreter | OCaml : Oct – Dec 2024

- Built a stack-based interpreter for an OCaml-like bytecode language supporting arithmetic, boolean logic, strings, control flow, and error handling.
- Implemented first-class functions and closures, including recursion, higher-order functions, and lexical environment capture.
- Added support for function calls, returns, and in/out parameters, correctly restoring stack and environment state across calls.

Memory Allocator | CNov – Dec 2024

- Developed a custom memory allocator in C, implementing malloc, free, calloc, and realloc.
- Emphasized multi-pool and bulk allocation strategies for efficient memory reuse.

Escape from UB (Horror Survival Game) | Unreal Engine : Nov – Dec 2025

- Designed and built a horror survival game inspired by Five Nights at Freddy's, featuring camera surveillance, enemy AI, and player resource management.
- Implemented core gameplay systems including AI pathing, timed enemy movement, jumpscare triggers, and fail-state logic to build player tension.
- Created interactive UI elements such as camera feeds, control panels, and status indicators.

Song Rater | Java : Feb – Mar 2024

- Built a Java application for rating songs, supporting adding, removing, and averaging ratings across multiple entries.
- Wrote comprehensive test cases covering standard and edge cases to validate functionality.

TECHNICAL SKILLS

Languages: Java, Python, C, OCaml, JavaScript, HTML/CSS

Tools & Technologies: MongoDB, WebSockets, bcrypt, Git, Unreal Engine, IntelliJ IDEA

Concepts: REST APIs, Backend Development, Systems Programming, Memory Management, Object-Oriented Programming, Data Structures & Algorithms